

Roots

Concepts

- **Roots** typically perform three main functions:
 - **Anchoring** the plant firmly to a substrate, providing stability for the shoot system.
 - **Absorbing** water and minerals from the soil.
 - **Producing hormones** (cytokinin and gibberellin) that regulate shoot growth and development.
- The cylindrical shape of roots is ideal for absorption (high surface-to-volume ratio) and for penetrating the soil matrix.
- Roots and shoots have an integrated growth relationship; the root system size is often balanced with the shoot system size.

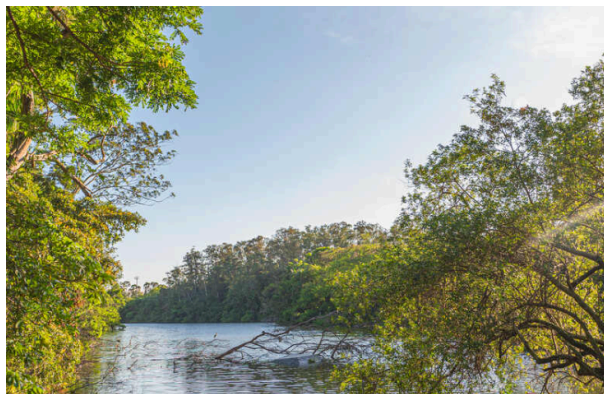


Figure 1: Roots play many roles in the life of a plant, but a common function is to provide anchorage and stability. The roots shown here are old and woody, but they were initiated when the tree was just a sapling; it grew from a seed that had landed on the side of a fallen tree, so the young roots had to grow over the trunk of the dead tree for more than a meter before they ever reached the soil. In several decades, the fallen tree will decompose completely, then the roots will be holding up the trunk of their tree, more than a meter above the ground with just air between them.

- Roots can persist long after the initial substrate (like a fallen log) has decomposed, providing structural support in unique ways.



Figure 2: The ratio of root system to shoot system is critically important; for each species, as the root system becomes heavier, so does the shoot system by a characteristic amount. However, the relationship differs for each species: Obviously for a given amount of turnip root there is relatively little shoot, but there is much more shoot for the same amount of pea root, and even more for cotton.

- There is a specific ratio of root mass to shoot mass for each species. As the shoot grows, the root system must also grow to support it.

External Structure of Roots

Organization of Root Systems

- **Taproot system:**
 - Develops from the **radicle** (embryonic root).
 - Consists of one main **taproot** and numerous smaller **lateral roots** (branch roots).
 - Common in most seed plants (gymnosperms and eudicots).
 - Can be specialized for storage (e.g., carrots, beets).



Figure 3: Dimensions of the Root System of Barley (*Hordeum vulgare*), 4 Weeks Old

- Root systems can be incredibly extensive, with total lengths reaching hundreds of meters or kilometers to provide sufficient surface area for absorption.



Figure 4: (A) Some taproots, such as a carrot, become extremely swollen and are much larger than the numerous lateral roots, whereas in other species, such as sunflower (B), the taproot is about the same size as the laterals. The important criterion is that the taproot develops from the root of the embryo, the radicle. (C) A fibrous root system, as in this winter wheat, consists of many roots, none of which is the radicle. Instead, all of the main roots are adventitious roots that originated in stem tissue.

- Taproots (A, B) originate from the radicle, while fibrous root systems (C) are composed of adventitious roots.

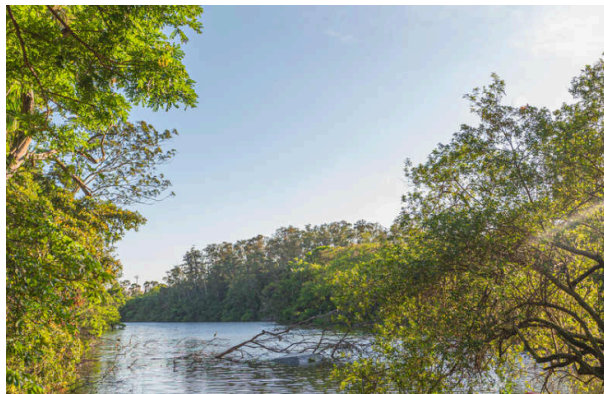


Figure 5: (A) The taproot of this tobacco seedling (*Nicotiana tabacum*) is definitely larger than the lateral roots, but more importantly, its development can be traced directly to the embryo. Relative size is not the critical factor; in many species, lateral roots are the ones that become enlarged. (B) The roots of radishes are obviously taproots.

- The taproot is the primary root derived from the embryo, distinct from lateral roots.
- **Fibrous root system:**
 - The radicle dies shortly after germination.
 - A mass of similarly sized roots arises from stem tissue.
 - These are **adventitious roots** (roots arising from non-root tissue).
 - Characteristic of **monocots**.



Figure 6: Corms of *Gladiolus*, like other monocots, have a fibrous root system. The radicle died shortly after germination; no root here has developed from the radicle. As the *Gladiolus* continues to grow, it will produce more roots, each independent of the others.

- In monocots like *Gladiolus*, the root system is fibrous and adventitious, not derived from a persistent radicle.
- **Secondary growth** and root systems:
 - Woody plants (eudicots/gymnosperms) increase conducting capacity via secondary growth, supporting a larger root and shoot system.
 - Monocots lack secondary growth; their conducting capacity is limited. They often rely on **adventitious roots** from rhizomes or stolons to supply new shoot growth.

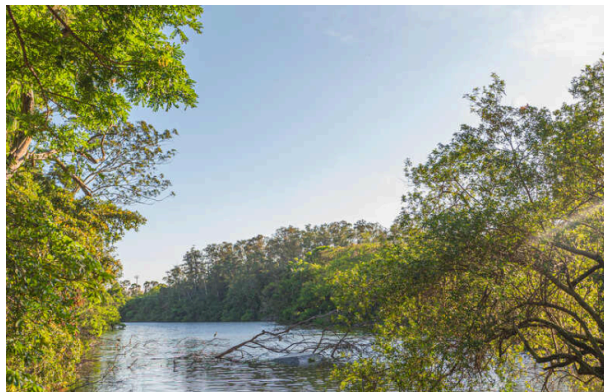


Figure 7: (A) A seedling of a woody plant has a few leaves and a small root system; the narrow trunk with a few vascular bundles can conduct water and nutrients between them. (B) An older woody plant has more leaves and a larger root system; the stem has more wood and bark, which increases its capacity to conduct water and sugar. Because most monocots do not undergo secondary growth, the stem of an older monocot is not wider (D) than that of a young plant (C), and it has no increased conducting capacity. Consequently, the old plant has no more leaves or roots than the young plant. (E) If a plant can produce adventitious roots, the bottleneck of the monocot stem does not matter. New roots originate near the aerial shoots and conduct directly into them, and little or no long-distance conduction occurs in the rhizome. No part of the monocot stem needs to conduct all of the water from all the roots to all the leaves and flowers, as does the trunk of a woody plant.

- Monocots overcome the lack of secondary growth by producing adventitious roots from new stem growth (E), whereas woody plants increase the capacity of existing stems and roots (B).

Structure of Individual Roots

- Roots grow by **localized growth** at the tip (apical meristem), not diffuse growth.
- Key regions of a root tip:
 - **Root cap:** Protects the apical meristem.
 - **Root apical meristem:** Zone of cell division.
 - **Zone of elongation:** Cells expand, pushing the root tip forward.
 - **Zone of maturation (Root hair zone):** Cells differentiate; root hairs form.



Figure 8: (A) The tip of a root consists of a root cap, the root apical meristem, a zone of elongation growth, and a region where root hairs are formed. (B) This radish seedling is only 3 days old and has abundant root hairs; those near the root tip are short because they are newer and younger than those near the shoot.

- Root hairs are restricted to the region of maturation, behind the zone of elongation, to prevent them from being sheared off during growth.

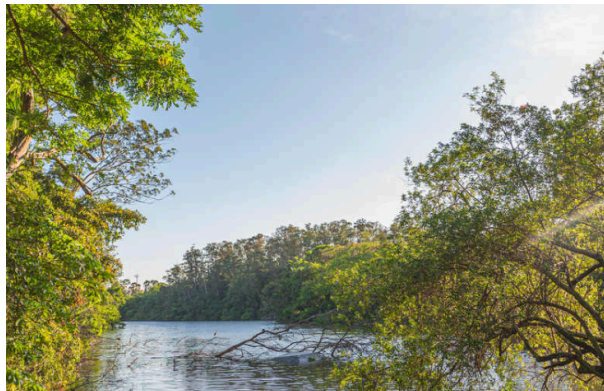


Figure 9: Although the exterior of a root appears rather uniform, several distinct zones of differentiation are present internally. Root hairs form only above the elongation zone, and the endodermis and first vascular tissues appear earlier than do root hairs.

- Internal differentiation (xylem, phloem, endodermis) begins in the zone of elongation, but full maturity occurs in the zone of maturation.

Internal Structure of Roots

Root Cap

- Protects the delicate **root apical meristem** as it pushes through soil.
- Cells are constantly worn away and replaced by the meristem.

- **Mucigel** secretion: Lubricates the root, aids in ion release from soil, and fosters beneficial bacteria.
- **Gravitropism**: Starch grains in columella cells settle to the bottom, detecting gravity.

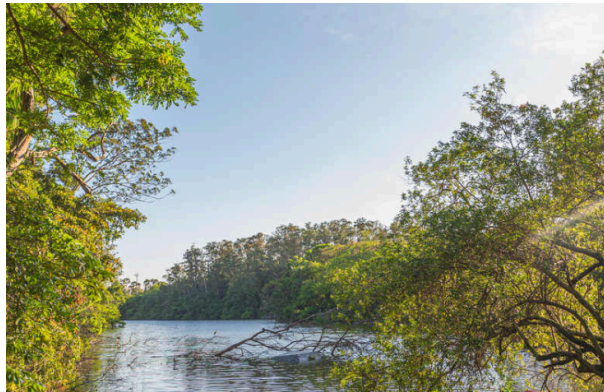


Figure 10: (A) The root cap is distinct from the root proper. The root apical meristem is located at the apex of the root proper but is buried under the root cap. Because no leaves, leaf traces, or branches are present, cells develop in an extremely orderly fashion in regular files. It is easy to see that these cells are derived directly from the meristem; thus, they make up primary tissues. (B) Screw pine (*Pandanus*) has giant aerial roots that allow us to see the root cap more easily than with ordinary tiny roots. The root cap cells die and are sloughed off.

- The root cap is a distinct structure covering the meristem. Its cells are arranged in files and are eventually sloughed off.

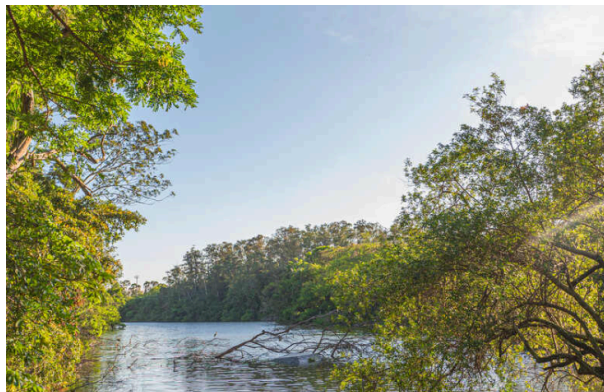


Figure 11: Much of a soil is composed of spaces between soil particles; in the uppermost 20 cm (uppermost dashed line) of a soil composed of sandy loam, 54% of the soil is pore space. Of this, 9% consists of extremely fine spaces less than $0.2\ \mu\text{m}$ wide (purple), 33% of spaces ranging between 0.2 and $60\ \mu\text{m}$ (green), and 12% of large spaces wider than $60\ \mu\text{m}$ (orange). At deeper levels, the soil is more compacted, containing only about 42% pore space (bottom of graph); the decrease is due mostly to a compaction of intermediate-sized spaces.

- Soil pores vary in size. Root hairs allow the plant to access water and minerals in fine pores that the main root body cannot enter.



Figure 12: In the root cap, cells in the central portion (A) divide so that the two daughter cells are aligned with the root and the root cap grows forward. On the edges (B), cells divide and expand in such a way that cells flow radially outward.

- Root cap growth involves central cells dividing transversely to push forward, while edge cells divide to expand radially.

Root Apical Meristem

- **Quiescent center:** A central region of mitotically inactive cells.
 - Acts as a reserve of healthy genetic material.
 - Can regenerate the meristem if it is damaged.

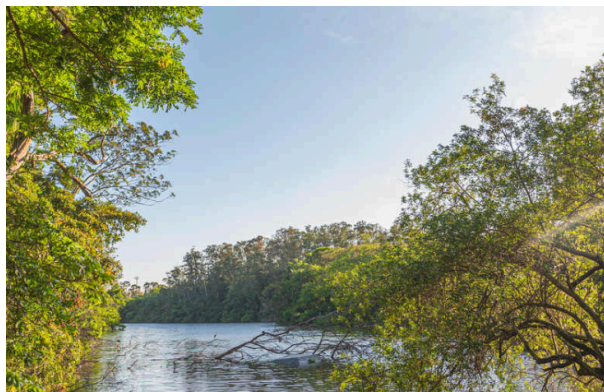


Figure 13: These roots were grown briefly in a solution of tritiated thymidine, a radioactive precursor of DNA. In cells that were undergoing the cell cycle, during S phase, the radioactive thymidine was incorporated into the nuclei. After a few hours, the root was killed, sliced into sections, and placed on photographic film in the dark. The radioactive nuclei caused black spots to form in the film next to the nuclei. The slide was then given a brief exposure to light so that the outlines of the cells and nuclei would be faintly visible (too much light would obscure the radioactivity-induced black spots). The quiescent center is the region where no nuclei became radioactive. Apparently no cell in the region passed through S phase while tritiated thymidine was available—these cells were in cell cycle arrest.

- The quiescent center is visible as a region where nuclei do not incorporate radioactive thymidine, indicating a lack of DNA synthesis (cell division).

Zone of Elongation

- Cells enlarge, primarily in length.
- Differentiation begins:

- **Protoderm** → Epidermis.
- **Provascular tissue** → Primary xylem and phloem.
- **Ground tissue** → Cortex.
- Tissues are permeable; minerals can enter via **apoplast**.

Zone of Maturation / Root Hair Zone

- **Root hairs:** Unicellular extensions of epidermal cells.
 - Greatly increase surface area for absorption.
 - Short-lived (days).
- **Cortex:**
 - Transfers minerals to vascular tissue.
 - Can function as **aerenchyma** (air storage) in some plants.
- **Endodermis:**
 - Innermost layer of the cortex.
 - **Casparian strips:** Bands of **suberin** and **lignin** on radial and transverse walls.
 - Makes cell walls impermeable to water and ions.
 - Forces all substances to pass through the **plasma membrane** (symplast) to enter the vascular cylinder, allowing the plant to control uptake.

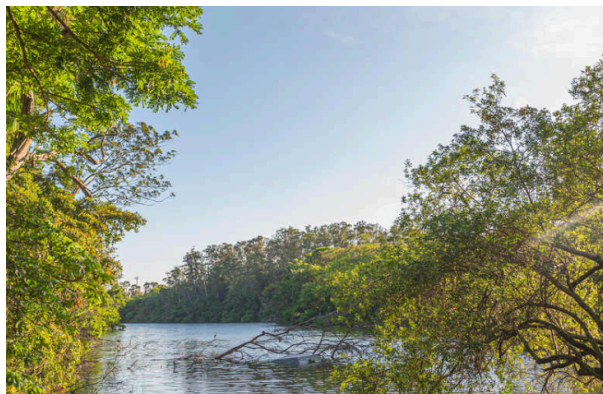


Figure 14: The cortex may be broad in many roots, and its interior boundary is the endodermis.

- The cortex occupies the space between the epidermis and the vascular cylinder, bounded internally by the endodermis.



Figure 15: (A) The endodermis is a cylinder, one layer of cells thick, each with Casparian strips. The best analogy is a brick chimney: It is a cylinder, one layer thick. The cement is analogous to the Casparian strips. (B) No apoplastic transport occurs across the endodermis. Materials that have been moving apoplastically (1) are stopped and can proceed farther only if the endodermis plasma membrane accepts them (2). Symplastic transport (3) is not affected. Plasma membranes that are impermeable to a particular type of molecule prevent that molecule from crossing the endodermis. (C) The endodermis in this corn is relatively easy to see: In all cells, Casparian strips are visible as red lines on the radial walls. In a few cells, only Casparian strips are present, but in many, the radial walls and inner tangential walls have started to thicken; it is unusual for this development to start so early.

- The Casparian strip blocks apoplastic transport (1), forcing water and minerals to cross the plasma membrane (2) to enter the vascular tissue.

Mature Portions of the Root

• Vascular Cylinder (Stele):

- **Pericycle:** Layer of parenchyma cells just inside the endodermis; site of lateral root origin.
- **Xylem:** Usually a solid central mass (in eudicots) or strands (in monocots).
 - **Protoxylem:** Outer, first-formed xylem.
 - **Metaxylem:** Inner, later-formed xylem.
- **Phloem:** Strands alternating with xylem arms.
- No **pith** in most eudicot roots; pith-like ground tissue present in many monocot roots.



Figure 16: (A) Low-magnification view and (B) high-magnification view of a transverse section of a eudicot root (buttercup, *Ranunculus*) showing the broad cortex and the small set of vascular tissues. The xylem has three sets of protoxylem, the narrow cells at the tips of the arms (B). The central, larger xylem cells are metaxylem. Three masses of phloem are also present; this is a triarch root. The pericycle is the set of cells between the endodermis and the vascular cells. (C) Low-magnification view and (D) high-magnification view of a transverse section of a monocot root (greenbrier, *Smilax*). The vascular tissue consists of numerous large vessels and separate bundles of phloem. The endodermis is very easy to see.

- Eudicot roots (A, B) typically have a solid central xylem core. Monocot roots (C, D) often have a ring of vascular bundles surrounding a central pith.



Figure 17: The number of strands of protoxylem in a root varies from one species to another and among the roots of one plant. Generally wider, more robust roots have more protoxylem masses. Having two masses of protoxylem (and two of phloem) is diarch (A). Three is triarch (B), and four is tetrarch (C). These diagrams represent three separate roots, not young and older stages of a single root; a diarch root does not become triarch and then tetrarch as it ages.

- Roots are classified by the number of protoxylem poles: diarch (2), triarch (3), tetrarch (4), etc.

Origin and Development of Lateral Roots

- **Lateral roots** originate from the **pericycle** (endogenous origin).
- They grow through the cortex and epidermis to emerge.
- Vascular tissues of the lateral root connect to the parent root.



Figure 18: This young lateral root of willow (*Salix*) was initiated in the pericycle, next to a mass of protoxylem; its growth caused considerable damage to the parent root cortex, and it broke the surface open. Entry of pathogens is prevented by formation of wound bark around the tear it caused.

- Lateral roots must push through and destroy cortex and epidermis cells to emerge, unlike axillary buds in stems which are superficial.

Other Types of Roots and Root Modifications

Storage Roots

- Fleshy roots (e.g., carrots, beets, sweet potatoes) store large amounts of carbohydrates (starch).
- Provide energy for new growth after winter dormancy (biennials/perennials).
- More stable environment than above-ground storage.

Prop Roots

- **Adventitious roots** arising from the stem above ground.
- Provide extra stability and absorption.
- Common in monocots (corn, palms, screwpine) and some eudicots (banyan, mangroves).

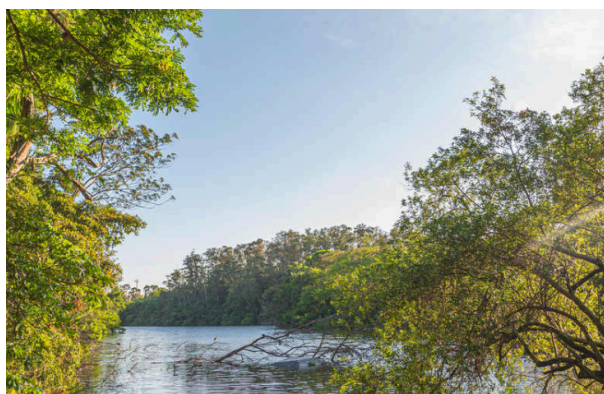


Figure 19: (A) Many monocots like this small palm (*Ptychosperma*) produce adventitious roots near the base of the stem. These provide both extra absorptive capacity and extra stability. (B and C) Screwpine (*Pandanus*) produces extremely long adventitious prop roots that not only stabilize the large, heavy trunk but also bring water and minerals into the stem. (D) In older plants of screwpine, even the branches produce adventitious roots. This water and mineral supply to the branches bypasses the trunk, where vascular bundles are already conducting at full capacity.

- Prop roots in monocots like *Pandanus* support the stem and can even support branches, bypassing the limited conduction of the main trunk.



Figure 20: (A) Banyan trees (eudicots) produce adventitious roots, which, as in monocots, provide increased support and absorptive capacity. Because the giant branches are supported along their entire length, they can become much larger and more extensive than branches that are supported only at the point of attachment to the trunk. (B) Plate-like buttress roots of *Ficus subcordata*. (C) These roots emerge from the basal portions of mangrove (*Rhizophora mangle*) in the Everglades National Park in Florida. Mangroves live in the intertidal zone, the band of seashore between high and low tide. At high tide, the roots are flooded with ocean water. This is a harsh environment because the seawater is so salty it can pull water out of most terrestrial plants and animals (that is why we cannot drink seawater), and the water should make the roots drown for lack of oxygen. Mangrove roots survive these conditions because their surface tissues keep seawater away from the delicate cells inside the root, and they have aerenchyma that lets oxygen diffuse into the root from the shoots.

- Banyan trees (A) and mangroves (C) use prop roots for support. Mangrove roots also have aerenchyma for oxygen transport in waterlogged soil. Buttress roots (B) provide stability for tall tropical trees.

Aerial Roots of Orchids

- Epiphytic orchids have roots adapted to dry air.
- **Velamen:** Multiple layers of dead epidermal cells that act as a sponge to absorb and retain water.



Figure 21: Aerial roots of orchids, which grow along the surface of tree bark, are covered by a white velamen that helps retain the water absorbed by the green root tip. These roots grow along the bark, not downward.

- The white velamen covers the aerial roots, preventing water loss and aiding absorption.

Contractile Roots

- Roots that shorten (contract) to pull the plant deeper into the soil.
- Important for bulbs, corms, and rhizomes (e.g., **Oxalis**, **Gladiolus**, **Crinum**) to maintain proper depth.



Figure 22: The roots of this hyacinth have contracted, causing their surfaces to become wrinkled.

- Wrinkled surface indicates contraction of the root, pulling the shoot downward.

Mycorrhizae

- Symbiotic association between roots and soil **fungi**.
- **Ectomycorrhizae**: Fungal hyphae surround the root and penetrate between cortical cells (common in woody plants).
- **Endomycorrhizae**: Hyphae penetrate cell walls of cortex cells (but not plasma membrane), forming **arbuscules** and **vesicles** (common in herbaceous plants).
- Benefit: Fungi greatly increase phosphorus absorption; plant provides sugars.



Figure 23: (A) Diagram of endomycorrhizae showing hyphae penetrating the root cortex and forming arbuscules and vesicles. (B) Root cells filled with fungal hyphae in the form of arbuscules. You can tell that this is a symbiotic fungus rather than a pathogenic one because the plant cells are healthy, with large normal nuclei. (C) Roots of beech encased in ectomycorrhizae. The presence of the fungus stimulates the roots to be short and broad. (D) Transverse section of an ectomycorrhizal root showing that the fungal filaments are restricted to just the outermost part of the root; they do not penetrate the cortex.

- Endomycorrhizae (A, B) penetrate cells to exchange nutrients. Ectomycorrhizae (C, D) form a sheath around the root.

Root Nodules and Nitrogen Fixation

- Symbiosis with nitrogen-fixing bacteria (e.g., **Rhizobium** in legumes).
- Bacteria convert atmospheric nitrogen (N_2) into ammonia (NH_3) or amino acids.
- **Root nodules**: Specialized structures formed by cortical cell division where bacteria live as **bacteroids**.
- **Leghemoglobin**: Plant protein that binds oxygen to protect nitrogenase enzyme.



Figure 24: (A) The nodules on these alfalfa roots contain bacteria capable of absorbing atmospheric nitrogen and converting it to ammonia, which the plant can use to make amino acids. (B) Bacteria in an infected root nodule cell of cowpea, *Vigna unguiculata*.

- Nodules house the nitrogen-fixing bacteria, providing them with a low-oxygen environment and sugars in exchange for nitrogen.

Haustorial Roots

- Modified roots of **parasitic plants** (e.g., mistletoe, dodder).

- **Haustoria:** Penetrate host tissues to connect with xylem (and sometimes phloem) to steal water and nutrients.



Figure 25: Transverse section of a branch of juniper being attacked by the haustorium (modified root) of a mistletoe (*Phoradendron*). The wood of the host juniper has been stained blue; the bark is tan. The haustorium is able to draw water and nutrients from the host vascular tissue.

- The haustorium penetrates the host's wood (xylem) to divert water and minerals to the parasite.

Roots of Strangler Figs

- Begin as epiphytes; roots grow down the host trunk to the soil.
- Roots fuse and encircle the host, eventually preventing its growth or shading it out.
- The fig becomes a free-standing tree with a hollow center where the host was.

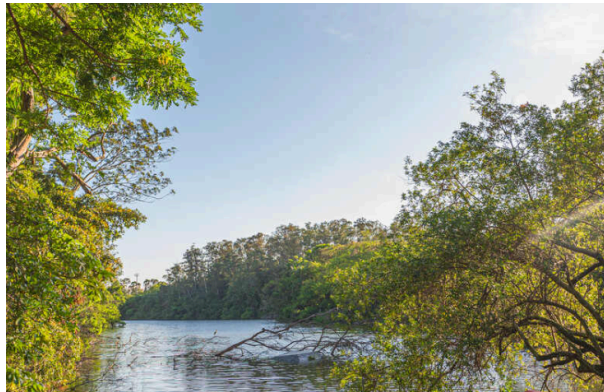


Figure 26: This tree in the Everglades National Park is encased in the roots of a strangler fig. The large “branch” of the fig at the upper left is actually the trunk of the fig, which is the point where the seed germinated. All portions below that are roots of the strangler fig. It would be easy to mistakenly assume the fig is a vine and the parts encasing the host tree are stems of the fig, but a close examination of the anatomy shows these are roots not branches. If this were really a vine, we would expect to see leafy branches emerging from many areas in this photo.

- Strangler fig roots form a mesh around the host tree, eventually providing structural support for the fig itself.